

Randomness Extraction from Unpredictable Random-Oracle Sources: Two Leftover-Hash-Lemma Conjectures

Setting. Fix nonempty finite sets $\mathcal{K}$ (seeds), $\mathcal{D}$ (inputs), $\mathcal{R}$ (outputs), and write $K := |\mathcal{K}|$, $D := |\mathcal{D}|$, $R := |\mathcal{R}|$. Let $\text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$ be the set of all functions $\mathcal{K} \times \mathcal{D} \rightarrow \mathcal{R}$, let $\text{SD}(\cdot, \cdot)$ denote statistical distance and $U_{\mathcal{R}}$ the uniform distribution on $\mathcal{R}$. The source S , the predictor P and the distinguisher D of Figure 1 are computationally unbounded and receive the *entire* function table of H as an explicit input. The two extraction games differ only in whether the seed sd is kept secret from D or given to it.

Game Pred_P^S $H \xleftarrow{\\$} \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$ $(x, z) \xleftarrow{\\$} S(H)$ $x' \xleftarrow{\\$} P(H, z)$ return $(x = x')$	Game Ext_D^S (secret seed) $H \xleftarrow{\\$} \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$ $(x, z) \xleftarrow{\\$} S(H); \quad sd \xleftarrow{\\$} \mathcal{K}$ $y_0 \leftarrow H(sd, x); \quad y_1 \xleftarrow{\\$} \mathcal{R}$ $b \xleftarrow{\\$} \{0, 1\}$ $b' \xleftarrow{\\$} D(H, y_b, z)$ return $(b = b')$	Game Ext-pub_D^S (public seed) $H \xleftarrow{\\$} \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$ $(x, z) \xleftarrow{\\$} S(H); \quad sd \xleftarrow{\\$} \mathcal{K}$ $y_0 \leftarrow H(sd, x); \quad y_1 \xleftarrow{\\$} \mathcal{R}$ $b \xleftarrow{\\$} \{0, 1\}$ $b' \xleftarrow{\\$} D(H, sd, y_b, z)$ return $(b = b')$
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Figure 1: Prediction game (left) and the two extraction games: secret seed (middle) and public seed (right). The first argument of S , P and D is the *full function table* of H ; the games Ext and Ext-pub differ only in whether D receives sd .

Definition 1. Set $\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}, S}^{\text{pred}}(P) := \Pr[\text{Pred}_P^S \Rightarrow 1]$ and

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext}}(S, D) := 2 \Pr[\text{Ext}_D^S \Rightarrow 1] - 1, \quad \text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext-pub}}(S, D) := 2 \Pr[\text{Ext-pub}_D^S \Rightarrow 1] - 1.$$

A source S is ϵ -unpredictable if $\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}, S}^{\text{pred}}(P) \leq \epsilon$ for every unbounded predictor P .

Conjecture 1 (Secret seed). *There is a universal constant $c > 0$, independent of $\mathcal{K}$, $\mathcal{D}$, $\mathcal{R}$ and ϵ , such that for all nonempty finite $\mathcal{K}, \mathcal{D}, \mathcal{R}$, all $\epsilon \in (0, 1]$, every ϵ -unpredictable source S and every unbounded distinguisher D ,*

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext}}(S, D) \leq \delta(\epsilon, K, D, R) := c \cdot \sqrt{\frac{\epsilon R + \log_2 D}{K}}.$$

Conjecture 2 (Public seed). *There is a universal constant $c > 0$, independent of $\mathcal{K}$, $\mathcal{D}$, $\mathcal{R}$ and ϵ , such that for all nonempty finite $\mathcal{K}, \mathcal{D}, \mathcal{R}$, all $\epsilon \in (0, 1]$, every ϵ -unpredictable source S and every unbounded distinguisher D ,*

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext-pub}}(S, D) \leq \delta_{\text{pub}}(\epsilon, K, D, R) := c \cdot \left(\sqrt{\epsilon R} + \sqrt{\frac{\log_2 D}{K}} \right).$$